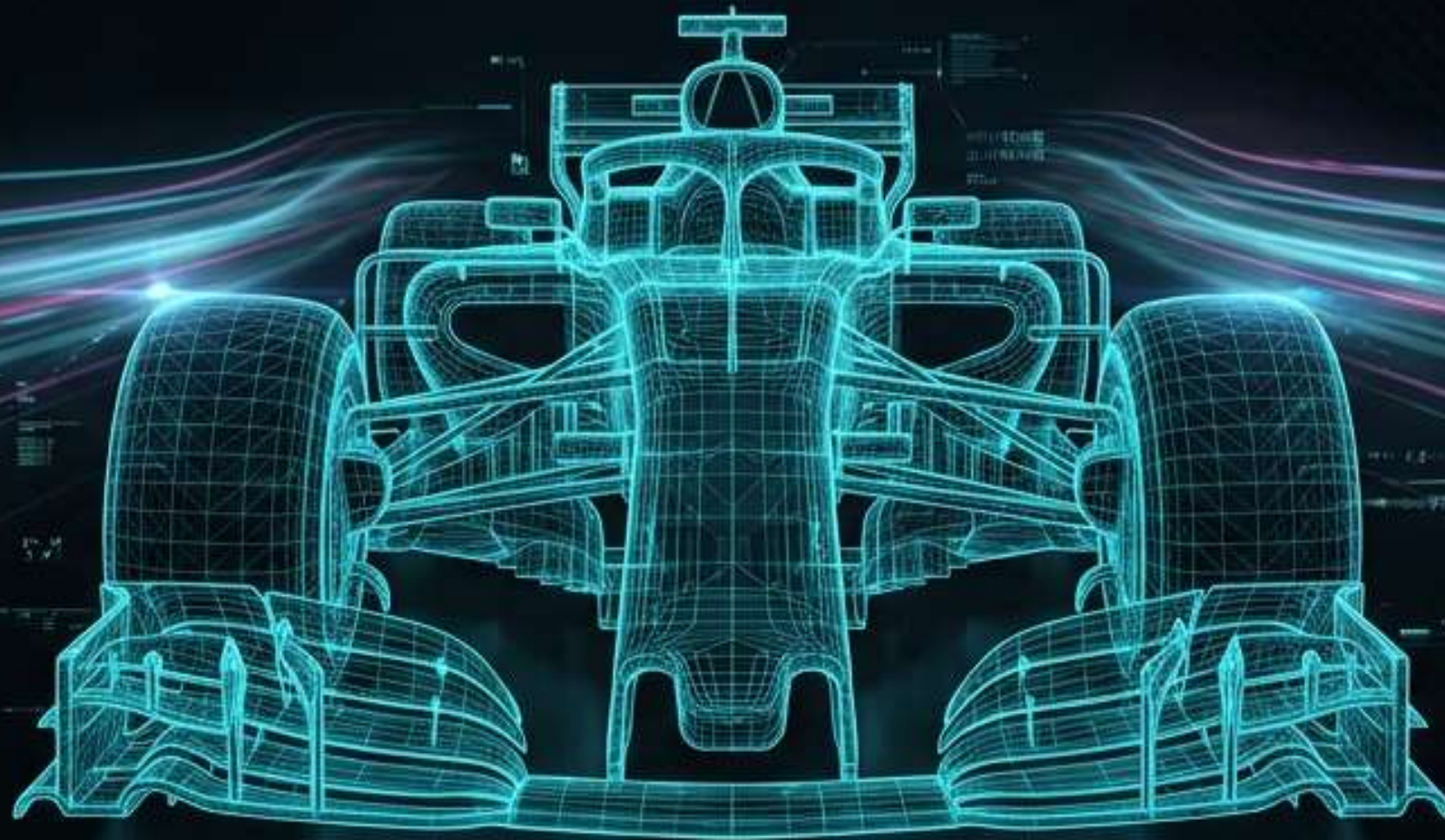


HYPER-SPEED TELEMETRY

A POLYGLOT DATA ARCHITECTURE FOR F1 IOT



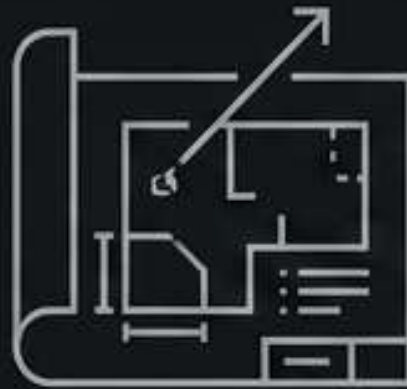
The IoT Challenge in Modern Motorsport

Scalability



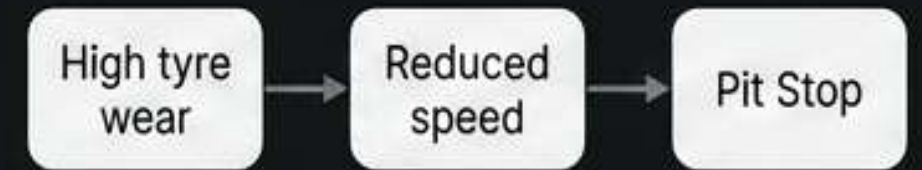
Terabytes of data per race weekend. Single databases fail under extreme write-throughput of hundreds of messages per second.

Flexibility



Fixed SQL schemas require costly, slow migrations as new telemetry sensors are rapidly introduced to the vehicle.

Relationships



Modeling complex event dependencies requires highly inefficient multi-table JOINS in traditional relational systems.

The Solution: Polyglot Persistence



MySQL

The Relational Layer

For structured aggregates
and lap-by-lap records.

MongoDB

The Document Layer

For flexible, schema-less
raw JSON logging and
debugging.

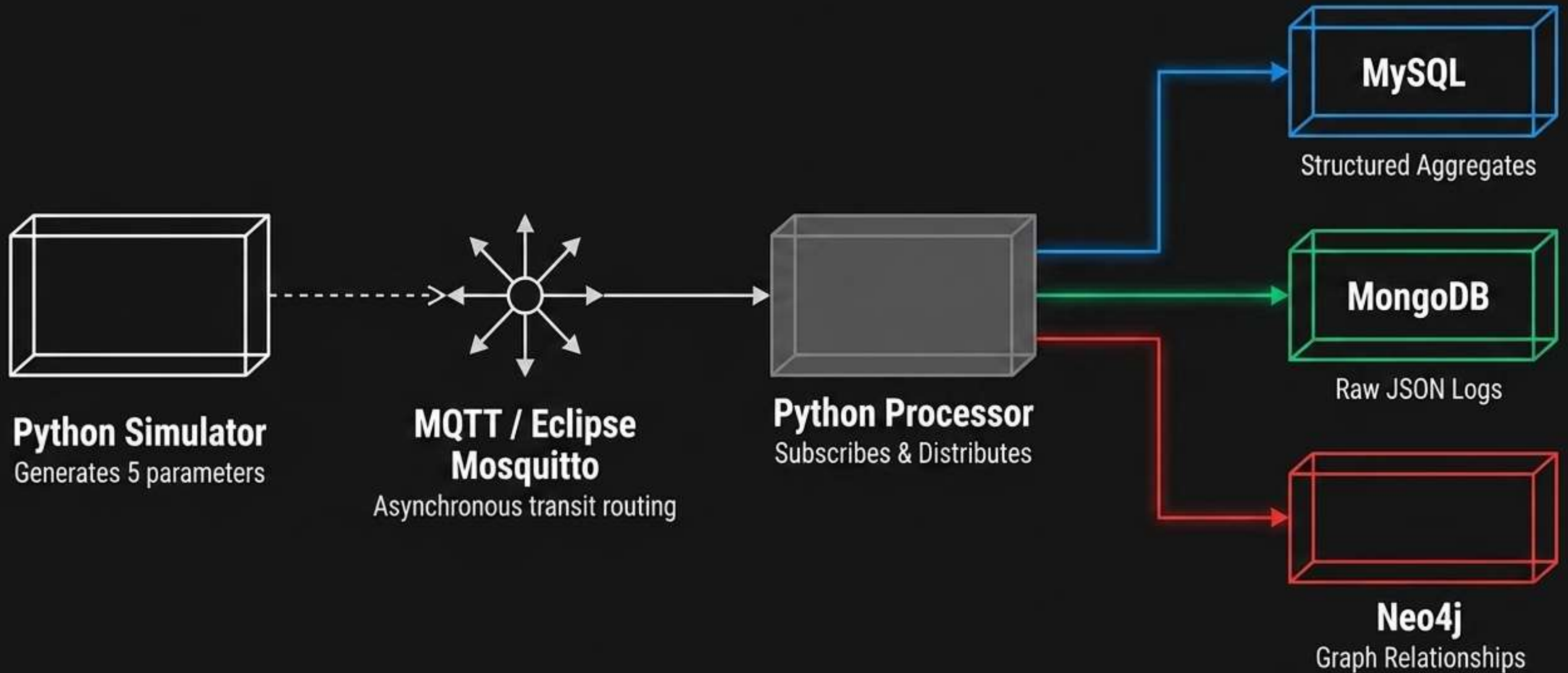
Neo4j

The Graph Layer

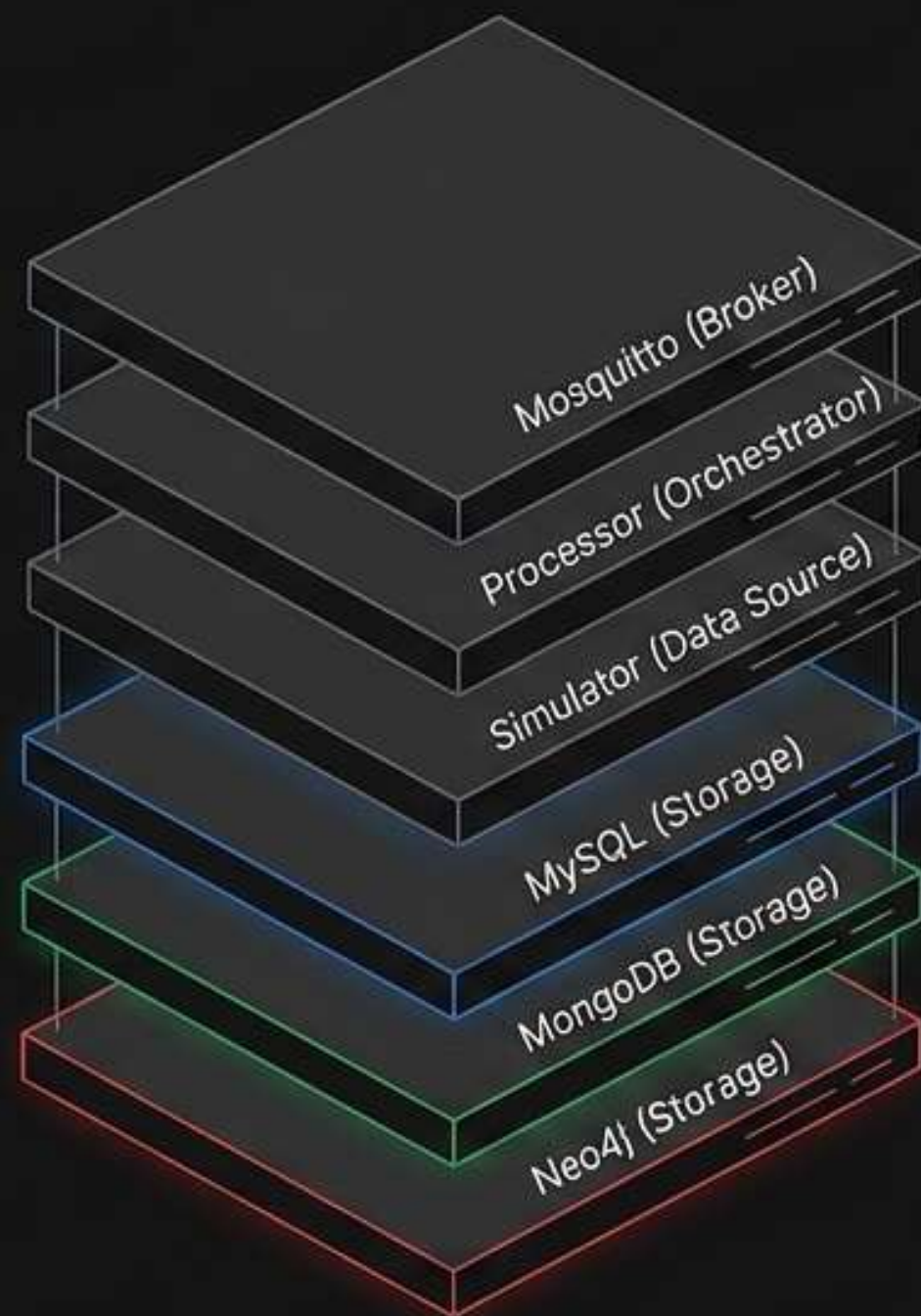
For modeling dependencies
between cars, laps, and tyre
wear events.

Racing data is massive, rapid, and varied. A multi-database architecture
applies the right storage paradigm to the right data problem.

Component Architecture & Data Flow



Reproducible Deployment via Docker



Unified Ecosystem

Using "docker-compose.yml", all 6 services deploy simultaneously on a shared internal network.

The processor connects securely to databases via service names (e.g., "mongodb:27017") rather than rigid IP addresses, completely eliminating environment inconsistencies.

Generation & Asynchronous Transit

The Payload

```
{  
  car: 'Ferrari_16',  
  lap: 58,  
  speed: 277,  
  fuel: 80.23,  
  tire_wear: 0.31  
}
```

The Transit



MQTT Protocol (Eclipse Mosquitto)

Publisher: Python Simulator

Broker: Mosquitto

Why MQTT?: Designed for constrained IoT environments. It decouples the publisher from the receiver, tolerates intermittent connectivity, and operates with massive bandwidth efficiency compared to standard HTTP.

The Orchestrator: Python Processor

```
def on_message(client, userdata, msg):  
    payload = msg.payload.decode()  
    data = json.loads(payload)  
    car = data['car']  
    lap = data['lap']
```

Single Responsibility Principle

The processor subscribes to the 'f1/telemetry' MQTT topic, intercepts the high-speed JSON messages, parses the telemetry parameters, and executes simultaneous writes to the three database endpoints.

Intelligence Layer

The Python simulator contains absolutely no complex logic. All processing, analytics, and storage delegation occur entirely within this centralized subscriber node.

Storage 1: Relational (MySQL)

lap	avg_speed	speed_smooth
70	258.31	263.94
69	272.27	267.64

- **Role:** Enforces a fixed schema to guarantee data consistency for core race metrics.
- **The Schema:** A single 'laps' table tracking car, lap, speed, fuel, and tyre wear.
- **Primary Use Case:** Aggregation functions. Ideal for computing lap-by-lap analytics or identifying the top 5 fastest laps.

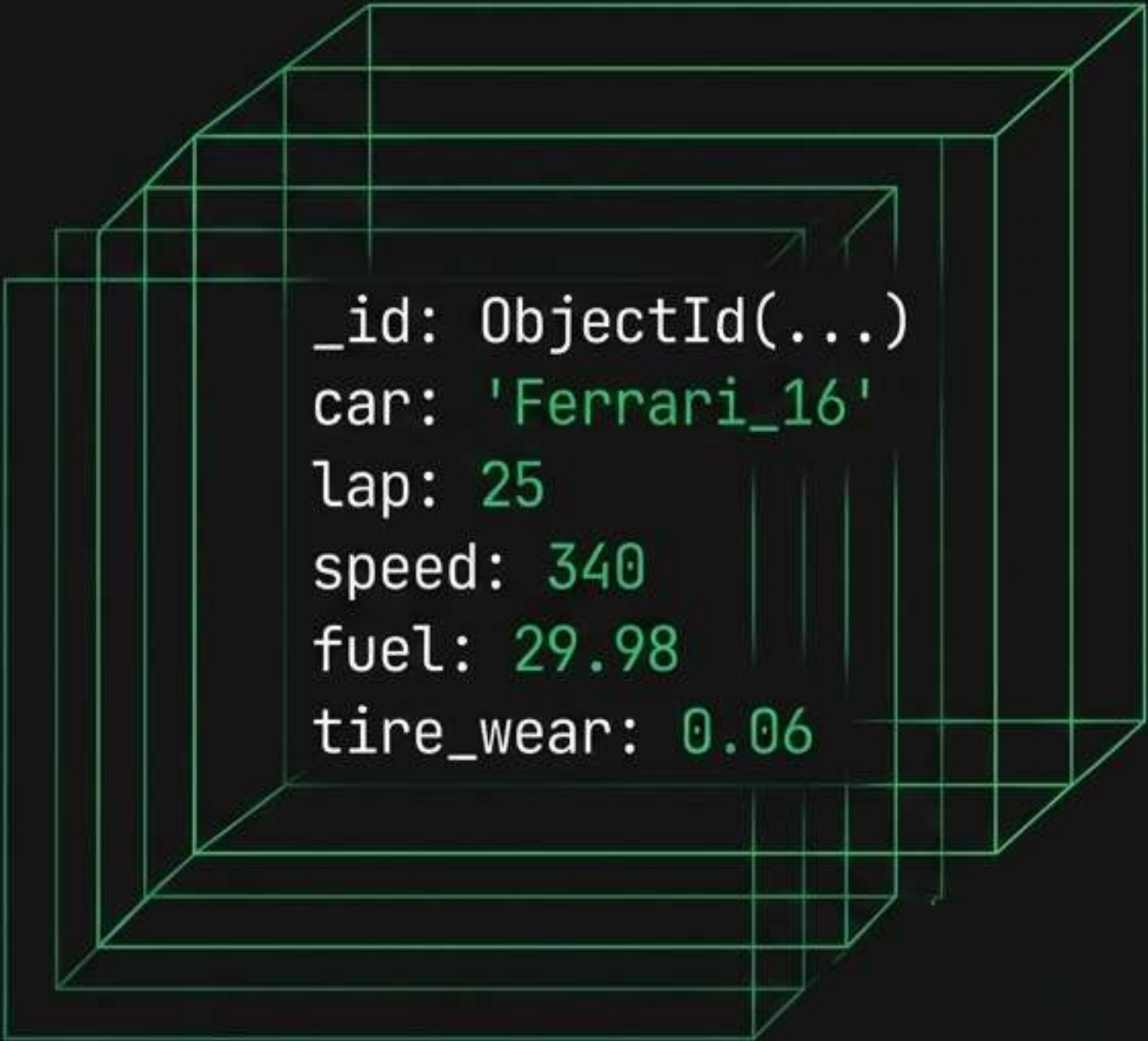
```
SELECT AVG(speed) FROM laps  
> Result: 261.63 km/h
```


Storage 2: Document (MongoDB)

Role: Acts as the unadulterated telemetry log. Stores incoming JSON messages exactly as received via 'pymongo'.

The Schema: Schema-less 'raw_data' collection.

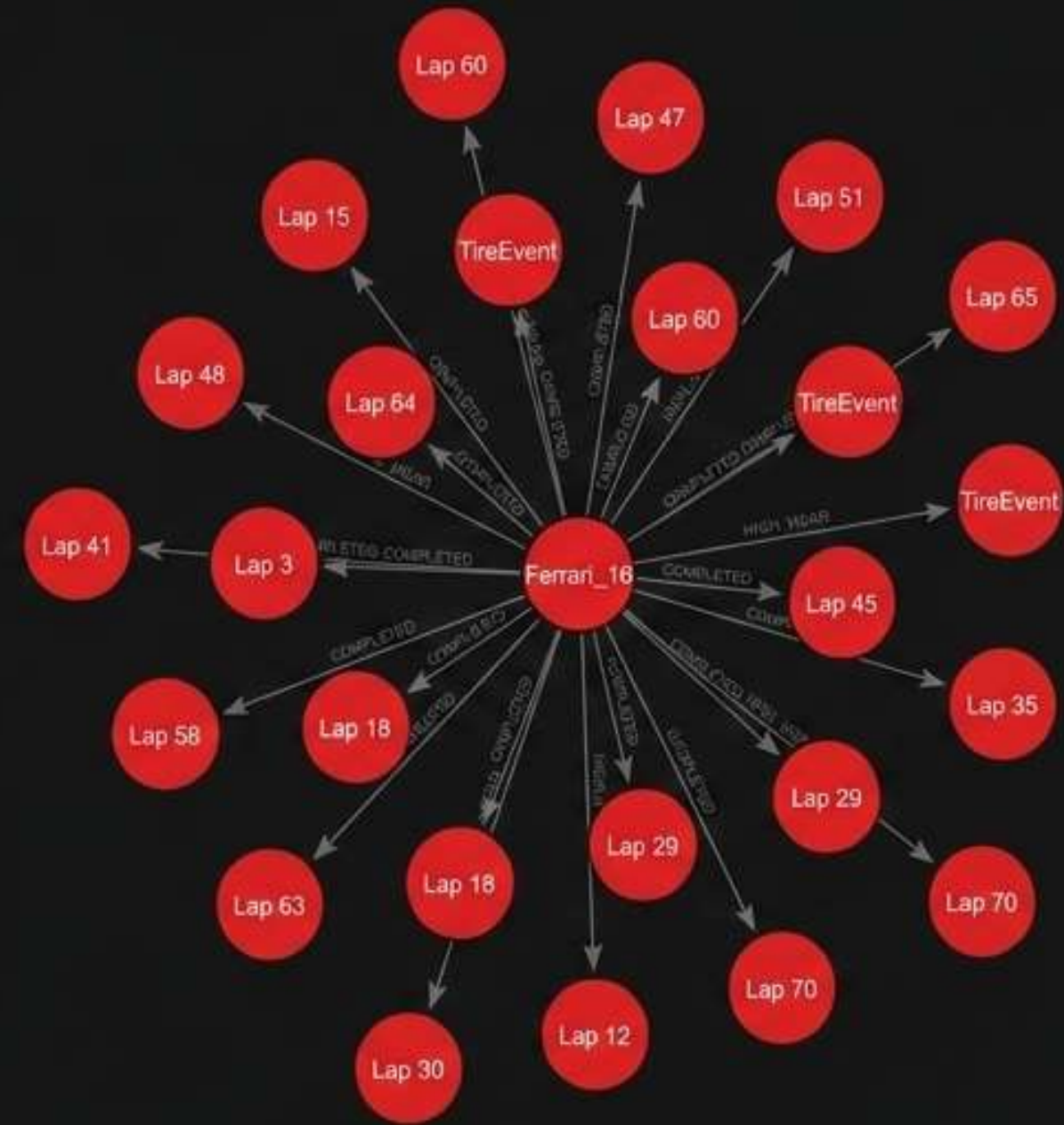
Primary Use Case: Rapid ingestion and ad-hoc debugging. Allows the simulator to add new data fields (e.g., brake pressure) mid-race without requiring costly database migrations or experiencing any downtime.



```
{
  "_id": ObjectId(...),
  "car": 'Ferrari_16',
  "lap": 25,
  "speed": 340,
  "fuel": 29.98,
  "tire_wear": 0.06
}
```

Storage 3: Graph (Neo4j)

- **Role:** Elevates relationships to first-class citizens, circumventing the need for slow, complex SQL JOINS.
- **The Schema:** Nodes ('Car', 'Lap', 'TireEvent') connected strictly by relationships ('COMPLETED', 'HIGH_WEAR').
- **Primary Use Case:** Event sequence analysis. Instantly maps and counts complex relational queries.

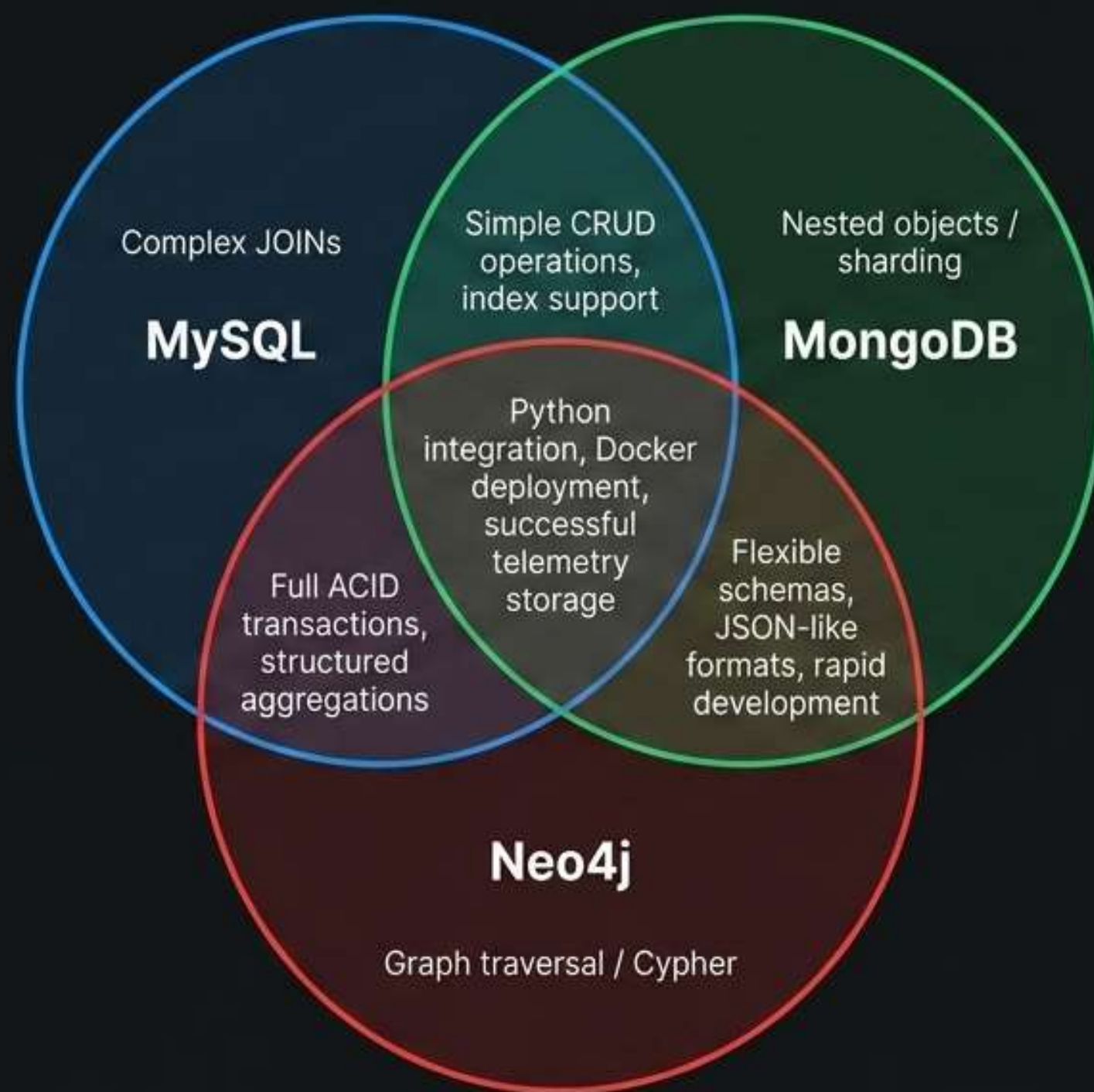


Show me the exact sequence of laps with increasing tyre wear.

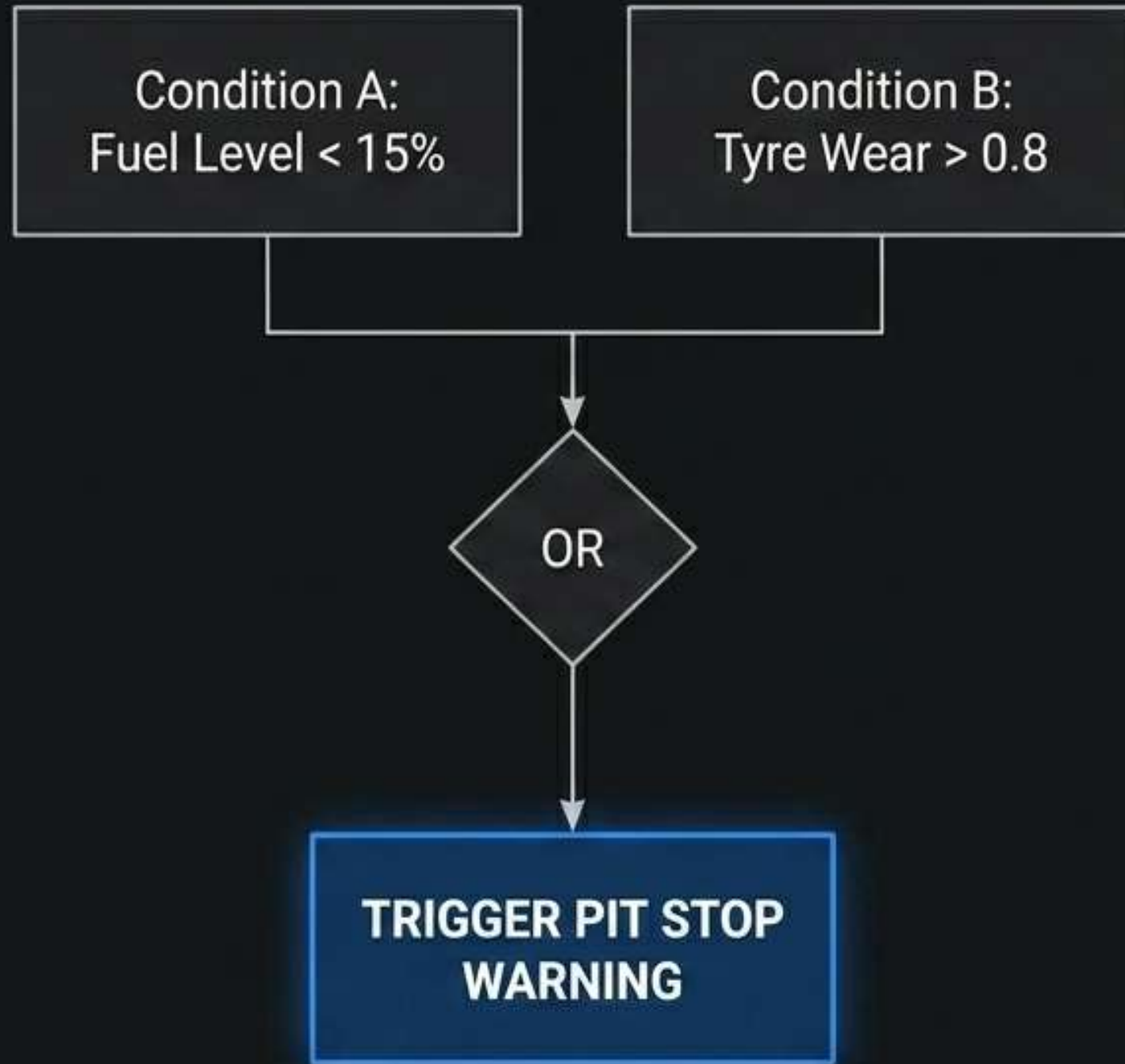
Comparative Database Evaluation

	MySQL	MongoDB	Neo4j
Data Model	Relational	Document	Graph
Schema	Fixed (migrations needed)	Flexible (schema-less)	Flexible Node Properties
Query Language	SQL	Aggregation Pipeline	Cypher
Performance Strengths	Fast aggregations & ACID compliance	High-volume raw inserts	Native, rapid relationship traversal

The Polyglot Synthesis



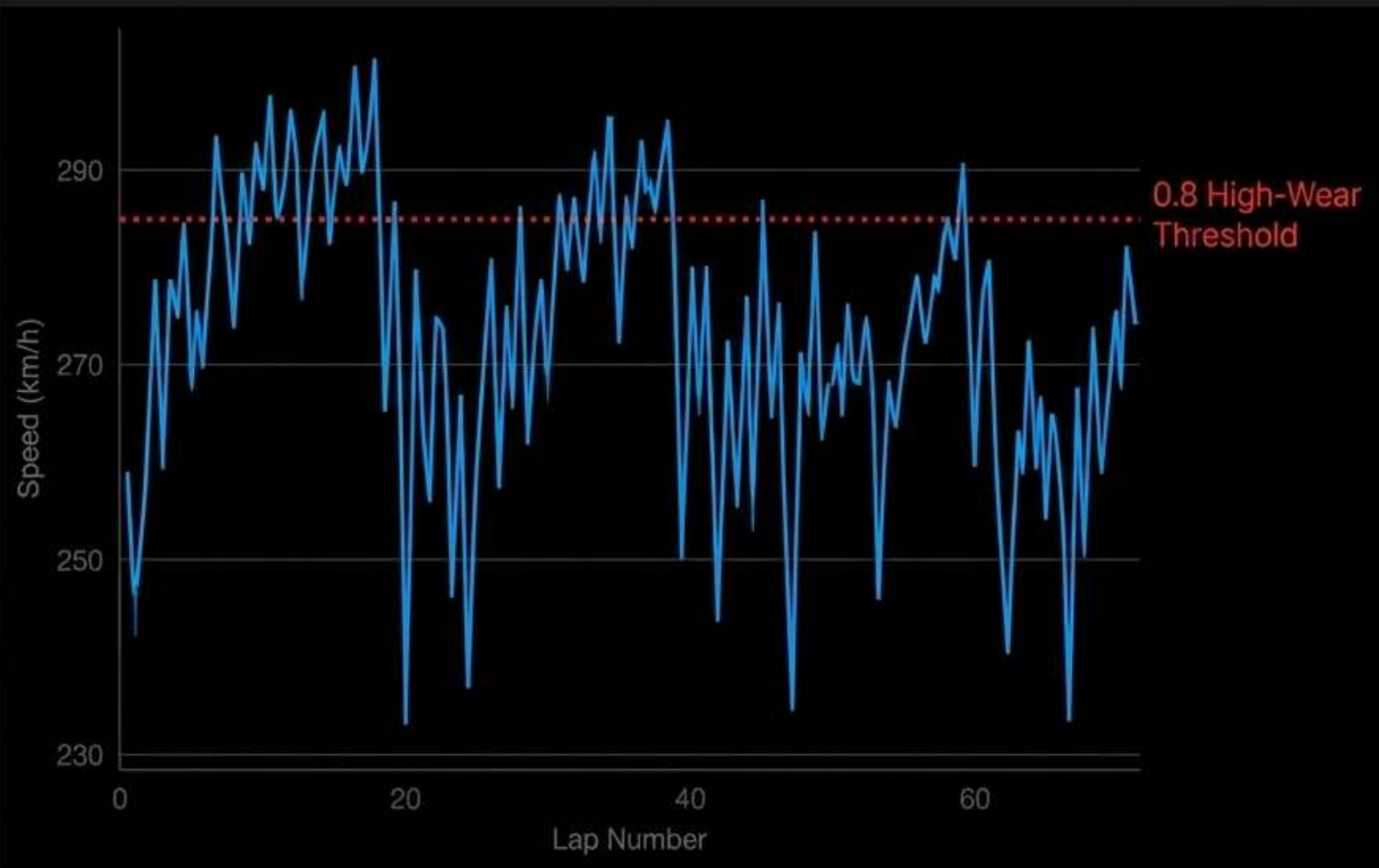
Real-Time Pit Stop Decision Support



```
[PIT STOP RECOMMENDED] car=Ferrari_16 fuel=9.59 wear=0.22  
[PIT STOP RECOMMENDED] car=Ferrari_16 fuel=7.79 wear=0.46
```

Results: Over an 1,100+ event test run, the rule-based predictor successfully triggered 84 times for critical fuel levels and 130 times for critical tyre wear with zero latency.

Telemetry Data Insights



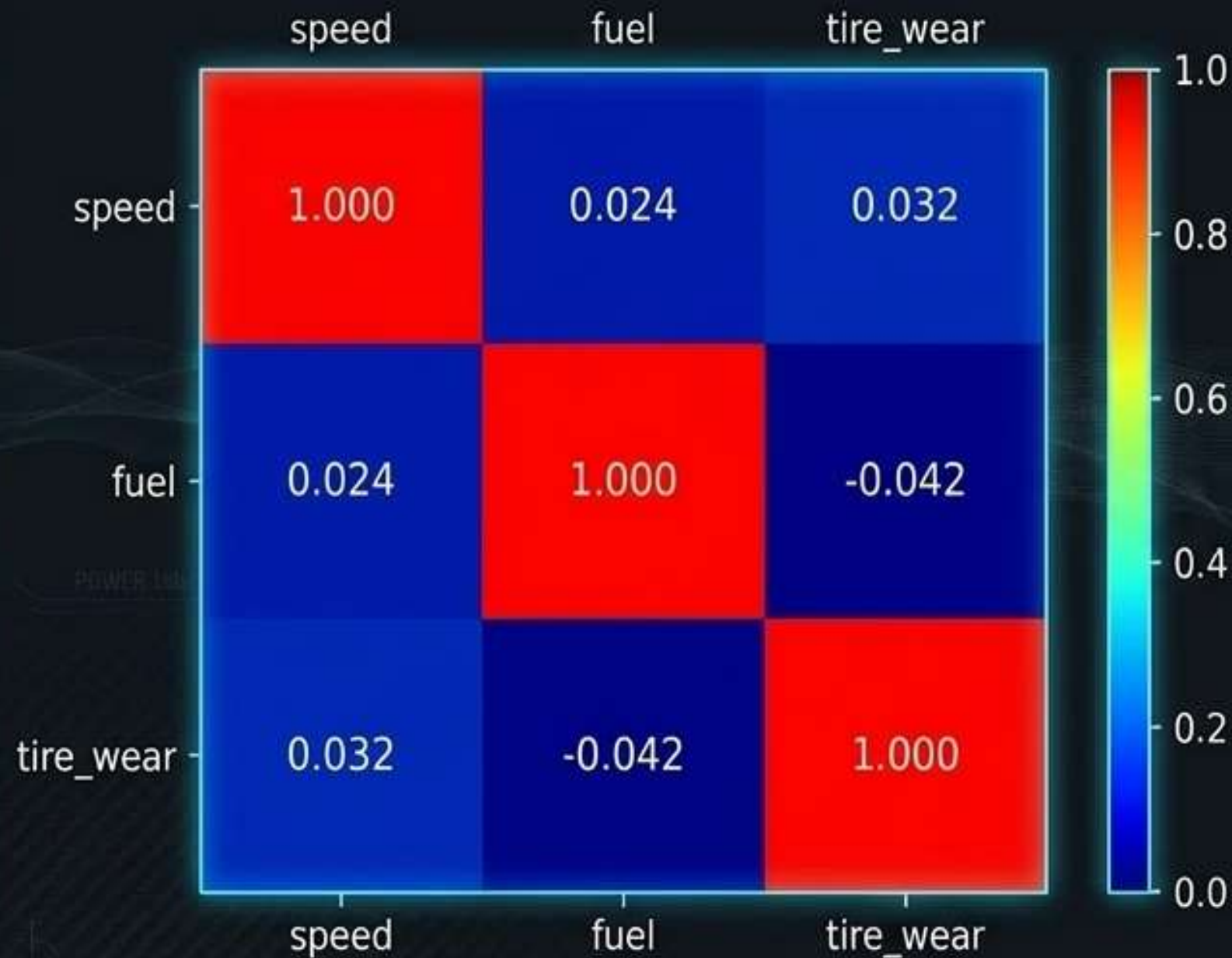
Insight

SQL analysis revealed that when tyre wear crosses the 0.8 threshold, the car's average speed drops by approximately 15 km/h.

Takeaway

The architecture not only logs data, but proves real-world racing physics: physical degradation directly correlates to measurable performance loss, objectively validating the Pit Stop Predictor's thresholds.

Telemetry Insights: The Cost of Wear



THE ANALYSIS:

Data extracted directly from the pipeline validates the real-world accuracy of the simulation.

THE CORRELATION:

A clear negative correlation between tire wear and vehicle speed.

THE CRITICAL THRESHOLD:

Once tire wear exceeds the 0.8 mark, average speed drops by -15 km/h.

CONCLUSION:

The data proves that critical tire degradation rapidly accelerates in later race stages, justifying the strict 0.8 pit-stop threshold.

Architecture Roadmap & Future Scalability

Machine Learning

Upgrading the rule-based pit predictor to a Random Forest model trained on historical MongoDB logs.

Live Integration

Ingesting UDP broadcasts from racing simulators (Assetto Corsa) or real CAN bus data.

Visual Dashboards

Connecting Grafana directly to the databases for live telemetry monitoring.

Enterprise Scaling

Replacing MQTT with Apache Kafka to handle massive telemetry workloads from entire fleets of cars.

Polyglot persistence is not just an architectural choice—it is a competitive necessity for modern motorsport.